COMP 2711
Discrete Math Tools for Computer Science
Mock Midterm Examination 04

Practice material – not an official examination

Time allowed: 120 minutes Total: 100 points

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This closed-book practice exam consists of 10 problems. You may use one A4 single-sided handwritten cheat sheet.
- Answer all problems in the spaces provided. You may also use the backs of pages and the draft pages. Each problem starts on a new page; some answers require only a few lines.
- Give a proof, justification, or counterexample when requested. State any additional notation you introduce.
- Answers may be left in exact form using binomial coefficients, factorials, and $C(n,k)$ or $P(n,k)$. For example, $\binom{5}{3} + \binom{4}{2}$ is acceptable. Avoid nonstandard notation such as nCk or nPk.
- The separate solutions document contains derivations and marking notes. Attempt the exam before consulting it.

Problems	1	2	3	4	5	6	7	8	9	10	Total
Points	10	13	12	16	10	9	7	6	7	10	100
Score											

Problem 1: Propositional logic and proofs [10 points]

(a) **[3 points]** Classify each formula as a tautology, contradiction, or contingency, with brief justification: (i) $(p \wedge q) \rightarrow p$; (ii) $(p \rightarrow q) \wedge p \wedge \neg q$; (iii) $p \leftrightarrow \neg q$. Each is worth 1 point.

(b) **[2 points]** Each of A and B is either a knight (always truthful) or a knave (always lying). A says, *B is a knight.* B says, *We are of different types.* Determine both types and justify.

(c) **[5 points]** From $\neg t \wedge u$, $r \rightarrow t$, $\neg r \rightarrow s$, and $s \rightarrow v$, derive $v \wedge u$ using named rules of inference.

Problem 2: Predicate logic [13 points]

(a) **[4 points]** In $(\forall x(P(x,y) \to \exists y\, R(x,y))) \wedge Q(y)$, identify every free and bound variable occurrence and each quantifier's scope (3 points). Rename the inner bound variable to remove the repeated name, preserving meaning (1 point).

(b) **[4 points]** All domains are nonempty. Rewrite $(\forall x P(x)) \to (\exists y Q(y))$ with all quantifiers at the front (2 points). Show it is not equivalent to $\forall x \exists y(P(x) \to Q(y))$ by a countermodel (2 points).

(c) **[5 points]** Prove formally that $\forall x \forall y(R(x, y) \rightarrow S(x, y))$ and $\exists x \forall y R(x, y)$ imply $\exists x \forall y S(x, y)$. Explain the use of a particular witness and an arbitrary variable.

Problem 3: Sets and functions [12 points]

(a) **[2 points]** Let $A = \{0, 1\}$, $B = \{a\}$. List $A \times B$ and give a relation from A to B which is not a function $A \to B$.

(b) **[3 points]** Use the Schröder–Bernstein theorem to show $|(0, 1)| = |[0, 1]|$. State the theorem and exhibit both injections; do not prove the theorem.

(c) **[7 points]** Write T or F for each statement. No justification is required. Each item is worth 1 point.

(i) $A \cap (A \cup B) = A$ for all sets A, B.

(ii) $U \setminus (A \cup B) = (U \setminus A) \cup (U \setminus B)$ for all $A, B \subseteq U$.

(iii) If finite A, B are disjoint, then $|A \cup B| = |A| + |B|$.

(iv) $(a, b) = (b, a)$ for all objects a, b.

(v) If $A \subseteq B$, then $A \setminus B = \varnothing$.

(vi) A one-to-one correspondence means a function that is both injective and surjective.

(vii) An injection from A to B and one from B to A force $|A| = |B|$.

Problem 4: Short counting calculations [16 points]

Give exact answers. Factorials, powers, and binomial coefficients may be left unevaluated. No explanation is required unless requested.

(a) **[2 points]** Count subsets of $\{1, \ldots, 8\}$ that contain 1 and do not contain 2.

(b) **[2 points]** Count injective functions $f : \{1, 2, 3\} \to \{a, b, c, d, e\}$ with $f(1) \neq a$.

(c) **[2 points]** Count length-five palindromic strings over $\{0, 1, 2\}$ whose first symbol is nonzero. Repetition is allowed.

(d) **[2 points]** Count length-four strings over $\{A, B, 0, 1\}$ with exactly two digit positions. Repetition is allowed.

(e) **[2 points]** Draw a branching tree for bit strings of length four with no consecutive ones, and count its leaves.

(f) **[2 points]** From ten distinct files, assign two to labelled folder A, three to labelled folder B, and leave the other five outside. A specified file Z must not enter A. Within-folder order does not matter. Count the assignments.

(g) **[2 points]** Count nonnegative integer solutions to $x_1 + x_2 + x_3 + x_4 = 8$ in which x_1 is even.

(h) **[2 points]** A row of Pascal's triangle is $1, 4, 6, 4, 1$. Give the next row, stating how boundary and interior entries are obtained.

Problem 5: Combinatorial proof [10 points]

[**10 points**] For $1 \leq r \leq t \leq n$, prove combinatorially that

$$\sum_{j=t}^{n} \binom{j-1}{r-1} = \binom{n}{r} - \binom{t-1}{r}.$$

Problem 6: Inclusion-exclusion [9 points]

(a) **[6 points]** For $n \geq 2$, use inclusion–exclusion on fixed-point events to derive the number D_n of permutations of $\{1, \ldots, n\}$ with no fixed point.

(b) **[3 points]** All $n!$ permutations are equally likely. Express the probability of no fixed point, and state the constant it approaches as n grows. No limit proof is required.

Problem 7: Pigeonhole principle [7 points]

[7 points] Prove that placing $N \geq 1$ objects in $k \geq 1$ labelled boxes forces one box to contain at least $\lceil N/k \rceil$ objects. Then suppose 22 files are placed in eight folders, each holding at most four files. Prove that at least three folders hold at least three files.

Problem 8: A fixed batch of trials [6 points]

Four independent transmission attempts each succeed with probability 1/3. The sample space is the four-symbol success/failure strings, with product masses. Let X count successes; all four attempts are made.

(a) **[1 points]** State the distribution of X, including its support.

(b) **[1 points]** Find $P(X = 2)$.

(c) **[1 points]** Find $P(1 \leq X \leq 2)$.

(d) **[1 points]** Find $P(X \geq 1)$.

(e) **[2 points]** Use the binomial theorem to verify that the pmf sums to one.

Problem 9: Birthday collisions [7 points]

In a toy calendar there are six days. Each person independently has a uniformly distributed birthday on these six days. Outcomes for n people are ordered n-tuples of days.

(a) **[3 points]** Use sequential conditional probabilities to find the probability of a shared birthday among three people.

(b) **[4 points]** Find the smallest n giving collision probability greater than $1/2$, and the smallest n guaranteeing a collision. Justify both thresholds.

Problem 10: Average costs [10 points]

An array contains $n \geq 2$ distinct keys in a uniformly random order. Linear search compares the target with successive keys, including the successful comparison. In (a),(b), the target is a fixed key present in the array.

(a) **[2 points]** Give the distribution and mean of the number L of comparisons.

(b) **[2 points]** A machine charges two units per comparison and refunds one unit per search. Find the expected charge.

(c) **[2 points]** If the target is absent and all n keys are compared, find the mean and variance of the comparison count.

(d) **[4 points]** Insertion sort instead receives a uniformly random permutation of n distinct keys. Count only shifts of earlier keys one place right; do not count condition tests or final key placement. Derive the expected total shifts.

Draft Paper

Mock examination working space.

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